

Python Programming Tasks

Submitted by: Radhika Bai

Question No 1 - Pay Calculator

Write a Python program that asks the user to enter the number of hours worked and the hourly rate. If the employee works 40 hours or less, pay = hours x rate. If more than 40 hours, the first 40 hours are paid at the normal rate and overtime hours at 1.5 times the hourly rate. Display the total pay.

Program (by Radhika Bai)

```
# Question No 1: Total pay calculator with overtime
# Programmer: Radhika Bai

def calculate_pay(hours, rate):
    """Return total pay. Hours above 40 are paid at 1.5 x rate."""
    if hours <= 40:
        return hours * rate
    overtime_hours = hours - 40
    return (40 * rate) + (overtime_hours * rate * 1.5)

print("=== Employee Pay Calculator ===")
hours_worked = float(input("Enter number of hours worked: "))
hourly_rate = float(input("Enter hourly rate: "))

if hours_worked < 0 or hourly_rate < 0:
    print("Hours and rate cannot be negative.")
else:
    total_pay = calculate_pay(hours_worked, hourly_rate)
    print("\n--- Pay Details ---")
    print(f"Hours Worked : {hours_worked}")
    print(f"Hourly Rate : {hourly_rate}")
    if hours_worked > 40:
        print(f"Regular Pay : {40 * hourly_rate:.2f} (40 hours)")
        print(f"Overtime Pay : {(hours_worked - 40) * hourly_rate * 1.5:.2f} "
              f"{(hours_worked - 40) hours at 1.5x}")
    print(f"Total Pay : {total_pay:.2f}")
```

OUTPUT - Sample 1: overtime worked

```
=== Employee Pay Calculator ===
Enter number of hours worked: 45
Enter hourly rate: 12.5

--- Pay Details ---
Hours Worked : 45.0
Hourly Rate : 12.5
Regular Pay : 500.00 (40 hours)
Overtime Pay : 93.75 (5.0 hours at 1.5x)
Total Pay : 593.75
```

OUTPUT - Sample 2: no overtime

```
=== Employee Pay Calculator ===
Enter number of hours worked: 35
Enter hourly rate: 10

--- Pay Details ---
Hours Worked : 35.0
Hourly Rate  : 10.0
Total Pay   : 350.00
```

Question No 2 - Advanced Student Marksheet

Ask for the student's name and roll number, input marks for 5 subjects, calculate total and percentage, determine the grade (80-100 A+, 70-79 A, 60-69 B, 50-59 C, 40-49 D, below 40 F), check pass/fail, declare Fail if any subject is below 40 regardless of overall percentage, and display a complete marksheet.

Program (by Radhika Bai)

```
# Question No 2: Advanced Student Marksheet
# Programmer: Radhika Bai

def get_grade(percentage):
    if percentage >= 80:
        return "A+"
    elif percentage >= 70:
        return "A"
    elif percentage >= 60:
        return "B"
    elif percentage >= 50:
        return "C"
    elif percentage >= 40:
        return "D"
    else:
        return "F"

def read_marks(subject_no):
    """Keep asking until a valid mark (0-100) is entered."""
    while True:
        try:
            m = float(input(f"Enter marks for Subject {subject_no} (out of 100): "))
            if 0 <= m <= 100:
                return m
            print(" Marks must be between 0 and 100.")
        except ValueError:
            print(" Please enter a valid number.")

name = input("Enter student's name: ")
roll_no = input("Enter roll number: ")

NUM_SUBJECTS = 5
marks = [read_marks(i + 1) for i in range(NUM_SUBJECTS)]

total = sum(marks)
percentage = (total / (NUM_SUBJECTS * 100)) * 100
grade = get_grade(percentage)
```

```

# Fail if overall below 40% OR any subject below 40
failed_subjects = [i + 1 for i, m in enumerate(marks) if m < 40]
if percentage < 40 or failed_subjects:
    result = "FAIL"
else:
    result = "PASS"

print("\n" + "=" * 40)
print("          STUDENT MARKSHEET")
print("=" * 40)
print(f"Name          : {name}")
print(f"Roll No.       : {roll_no}")
print("-" * 40)
print(f"{'Subject':<12}{'Marks':>10}{'Status':>12}")
for i, m in enumerate(marks, start=1):
    status = "Pass" if m >= 40 else "Fail"
    print(f"{'Subject ' + str(i):<12}{m:>10.1f}{status:>12}")
print("-" * 40)
print(f"Total Marks : {total:.1f} / {NUM_SUBJECTS * 100}")
print(f"Percentage  : {percentage:.2f}%")
print(f"Grade       : {grade}")
print(f"Result      : {result}")
if failed_subjects and percentage >= 40:
    print("Note       : Failed because marks are below 40 in "
          "Subject(s) " + ", ".join(map(str, failed_subjects)))
print("=" * 40)

```

OUTPUT - Sample 1: student passes

```

Enter student's name: Radhika Bai
Enter roll number: 104
Enter marks for Subject 1 (out of 100): 85
Enter marks for Subject 2 (out of 100): 78
Enter marks for Subject 3 (out of 100): 92
Enter marks for Subject 4 (out of 100): 88
Enter marks for Subject 5 (out of 100): 81

```

```

=====
                        STUDENT MARKSHEET
=====
Name          : Radhika Bai
Roll No.     : 104
-----
Subject       Marks       Status
Subject 1     85.0        Pass
Subject 2     78.0        Pass
Subject 3     92.0        Pass
Subject 4     88.0        Pass
Subject 5     81.0        Pass
-----
Total Marks : 424.0 / 500
Percentage  : 84.80%
Grade       : A+
Result      : PASS
=====

```

OUTPUT - Sample 2: fails (one subject below 40)

Enter student's name: Radhika Bai
Enter roll number: 104
Enter marks for Subject 1 (out of 100): 85
Enter marks for Subject 2 (out of 100): 35
Enter marks for Subject 3 (out of 100): 92
Enter marks for Subject 4 (out of 100): 88
Enter marks for Subject 5 (out of 100): 81

=====

STUDENT MARKSHEET

=====

Name : Radhika Bai

Roll No. : 104

Subject	Marks	Status
Subject 1	85.0	Pass
Subject 2	35.0	Fail
Subject 3	92.0	Pass
Subject 4	88.0	Pass
Subject 5	81.0	Pass

Total Marks : 381.0 / 500

Percentage : 76.20%

Grade : A

Result : FAIL

Note : Failed because marks are below 40 in Subject(s) 2

=====

Prepared and executed by **Radhika Bai**.